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Solid State Devices (EE3001)

Quiz#2 (Total Mark 15)

Date: 30th March, 2022

Time: 8 am to 8:50 am

Answer all questions. Final answers must be written in the space provided after each question. Marks will be deducted if proper units are not mentioned. All important steps to arrive at the final answer must be shown in the additional answer sheet.

Assume intrinsic carrier concentration (n_i) in Si is 10^{10} /cm^3 and kT = 26 meV at T = 300K. Bandgap and permittivity of Si can be assumed as 1.1 eV and $\epsilon_{Si}=10^{-12}$ F/cm, respectively.

Q1. The space charge density profile under equilibrium for an unknown semiconductor device structure is given in Fig. 1. Considering total charge in the region $x_1 < x < x_4$ to be zero, draw the nature of the corresponding electrostatic field variation with position. **[2]**

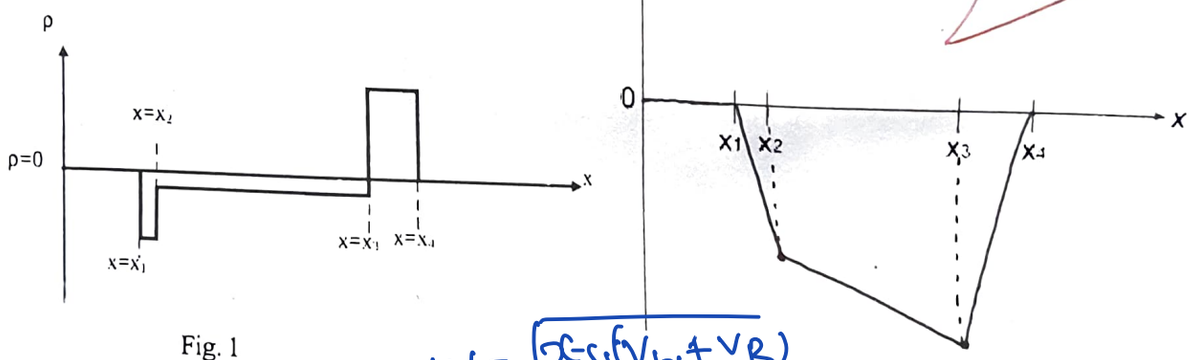


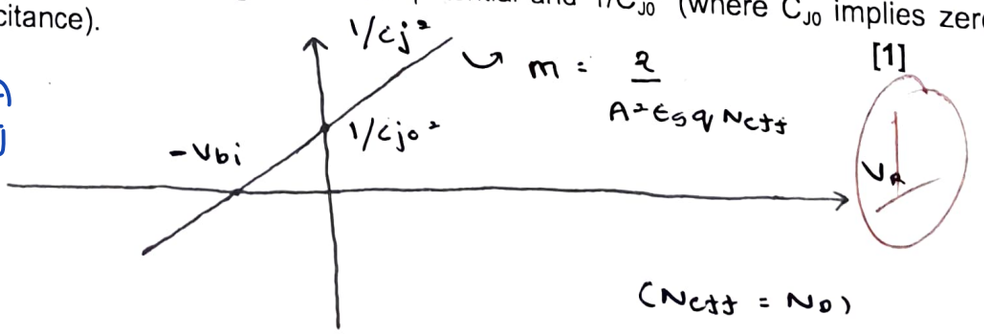
Fig. 1

2

$W = \sqrt{\frac{2\epsilon_{Si}(V_{bi}+V_R)}{qN_D}}$

Q2. (a) For an abrupt Si p$^+$n junction diode of cross-sectional area (A) equal to 1 cm^2, plot the nature of $1/C_J^2$ (C_J being the junction capacitance) versus applied reverse bias voltage (V_R). Clearly mark the points showing the built-in potential and $1/C_{J0}^2$ (where C_{J0} implies zero bias junction capacitance). **[1]**

$C_j = \frac{\epsilon_{Si}A}{W}$



$m = \frac{2}{A^2\epsilon_{Si}qN_{cff}}$

(b) If $1/C_{J0}^2 = 10^{16}$ /F^2 and the built in potential is 0.8 V,

doping concentration of the n-region is....$1a^{15}$...$1.cc$.....

[2]

2

$I_p = \frac{qAD_p p_{no}}{W_p}(e^{V/V_T} - 1) = \frac{qAD_p n_i^2}{W_n N_D}(e^{V/V_T} - 1)$

$I_p/I_n = \mu_p/\mu_n \times W_p/W_n \times N_A/N_D = 1/5$

Q3. In a p-n junction of uniform cross-section, the width of the neutral n-region (W_n) is much less than the hole diffusion length (L_p) and the width of the neutral p-region (W_p) is much less than the electron diffusion length (L_n). For this device, $W_p = 2W_n$. Also, the resistivities of the p-region and n-region are $\rho_p = 1$ Ω-cm and $\rho_n = 0.1$ Ω-cm respectively. If the device is forward biased so that the current flowing through the device is 3 mA,

$\sigma_p = q\mu_p N_A$

$\frac{\mu_n N_D}{\mu_p N_A} = 10$

(a) **the electron current component (I_n) is** ..... 2.5 .....

(b) **the hole current component (I_p) is** ..... 0.5 .....

[3]

$E_{max} = \frac{qN_{eff}W}{\varepsilon_{Si}}$

Q4. In a p^+nn^+ junction diode, the critical field at avalanche breakdown is 2×10^5 V/cm. The desired breakdown voltage is 30 V. Built-in potential can be neglected at breakdown.

If the n-region width is 2 μm, doping concentration of the n-region is ..... 3.125×10^{16}/cc **[3]**

Q5. A p^+n junction photodiode is illuminated with monochromatic light of **1 mW** optical power having $h\nu$ = **0.8 eV** (greater than the bandgap of the semiconductor), for which the absorption coefficient of the semiconductor material is 10^4 **/cm**.

1.28×10^{-19} J

$I(x) = I_0 e^{-\alpha x}$

(a) The fraction of the incident light absorbed till **2.5 μm** from the surface is ..... 0.9179 **[1]**

$1 - e^{-\alpha x}$

(b) If the light absorbed till **2.5 μm** from the surface only results in useful electron-hole pairs (EHPs), the additional light generated current (I_L) is ..... 2.299 mA ..... 1.15mA **[2]**

no of photons $= \frac{10^{-3}}{1.28 \times 10^{-19}} = 7.81 \times 10^{15}$

useful $= 0.9179 \times 7.81 \times 10^{15} = 7.17 \times 10^{15}$

$I = (1.6 \times 10^{-19})(7.17 \times 10^{15})$

Q6. Choose the correct answer: In their normal mode of operation,

(A) An LED and photodiode are forward biased whereas a Zener diode and a LASER are reverse biased

(B) a photodiode and a Zener diode are forward biased whereas a LASER and an LED are reverse biased

(C) a Zener diode and a LASER are forward biased whereas an LED and photodiode are reverse biased

(D) a LASER and an LED are forward biased whereas a photodiode and a Zener diode are reverse biased

The correct answer is ..... (D) .....

[1]